

# Derivation and Numerical Reproduction of Liang, Goldsman, Mayergoyz & Oldiges (1997): 2-D Deterministic SHE-Boltzmann MOSFET Simulation

*Complete report, Revision 2 — coupled, impact-ionization-in-BTE state*

Reproduction study (code commit 891e7c9; Python 3.13.14, NumPy 2.5.1, SciPy 1.18.0)

August 15, 2026

## Abstract

This report derives and numerically reproduces the deterministic spherical-harmonic (SHE) Boltzmann device solver of Liang *et al.*, *IEEE Trans. Electron Devices* **44**(2), 257–267 (1997). Revision 2 supersedes the earlier one-way/post-processed description: impact ionization is now *inside* the assembled electron collision operator (primary depletion + secondary redistribution), the self-consistent SHE $\leftrightarrow$ Poisson $\leftrightarrow$ hole-continuity loop is *demonstrated and observably converging*, global number- and terminal-current-conservation diagnostics are in place, and a physics-split preconditioner removes the previous solver bottleneck ( $\sim 10$  min  $\rightarrow \sim 15$  s on the  $40 \times 34$  problem). Established results: core transport ( $n$ ,  $|v|$ ,  $T_e$ ) is validated and spatially robust; the self-consistent feedback materially suppresses avalanche generation and drain current while leaving  $T_e$  comparatively stable; conservation holds across one-way, refined, and coupled runs. Open items are stated precisely: the impact-ionization magnitude is *not spatially converged* — a  $2 \times 2$  grid $\times$ coupling study drives  $G_{ii,\max}$  multiplicatively from  $1.35 \times 10^{27}$  to  $5.24 \times 10^{26} \text{ cm}^{-3}\text{s}^{-1}$  with no plateau, while an energy refinement to  $\Delta H = 6.25 \text{ meV}$  moves it only  $-2.4\%$  (the sensitivity is spatial, not energetic), so no converged value can be quoted; the  $T_e/G_{ii}$  peak offset is unresolved; and the device has now been recalibrated to  $V_{th} = 0.614 \text{ V}$  for the pending Liang overlays. The robust reproductions are the transport moments and the ( $\sim 37\%$ ) self-consistent feedback fraction. Every model element is classified as *paper-specified*, *reconstruction*, *surrogate*, or *not-yet-converged*.

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## 1 Overview and audit provenance

Liang *et al.* solve, self-consistently, three coupled equations on a 2-D MOSFET cross-section: Poisson, the electron BTE (first-order SHE), and hole continuity with an impact-ionization pair-generation source. This reproduction has passed through four external audit rounds and three code patches (Appendix B); the present revision documents the resulting coupled state.

**Honesty note.** Model substitutions carried throughout and classified in Table 1: the doping profile is a *reconstruction*; the band is a 1-parameter Kane *surrogate* (paper: 2-band Fiegna/Brunetti); the impact-ionization rate is a Keldysh *surrogate*; the momentum time  $\tau_1$  is a single calibrated surrogate (not a summed Table-I model); and the device carries an un-recalibrated  $V_{th} = 1.64$  V.

## Part I

# Method derivation and validated foundation

## 2 First-order SHE equation

With  $\mathbf{E} = -\nabla_{\mathbf{r}}\phi$ , expand  $f = f_0(\mathbf{r}, \varepsilon) + \hat{\mathbf{\Omega}} \cdot \mathbf{f}_1(\mathbf{r}, \varepsilon)$  (all  $\ell \geq 2$  discarded). Projection gives the current relation and the conservative augmented-space balance

$$\mathbf{f}_1 = -\tau_1 v (\nabla_{\mathbf{r}} f_0 - q \mathbf{E} \partial_{\varepsilon} f_0), \quad \nabla_{\mathbf{r}} \cdot \mathbf{j}_x + \partial_{\varepsilon} j_{\varepsilon} = Z \mathcal{Q}[f_0] + Z S_{\text{gen}}, \quad (1)$$

$\mathbf{j}_x = -ZD(\nabla_{\mathbf{r}} f_0 - q \mathbf{E} \partial_{\varepsilon} f_0)$ ,  $D = \frac{1}{3} v^2 \tau_1$ . In total energy  $H = \varepsilon - q\phi$  this becomes pure spatial diffusion at fixed  $H$ , the collision operator coupling  $H \leftrightarrow H \pm E_{op}$  locally in space, over the global mesh  $-q\phi_{max} \leq H \leq \varepsilon_{max} - q\phi_{min}$  with mask  $0 \leq H + q\phi \leq \varepsilon_{max}$ ;  $\varepsilon = 0$  reflecting,  $\varepsilon_{max}$  absorbing. The DOS-weighted acoustic operator  $\mathcal{Q}_{ac} = Z^{-1} \partial_{\varepsilon} [ZA(\partial_{\varepsilon} + 1/k_B T) f_0]$  (Scharfetter–Gummel) makes the collision term irreducible; it is number-conserving (defect  $7.6 \times 10^{-17}$ ) with the Maxwellian as its exact null.

## 3 Reconstructed device, taxonomy, foundation

The  $0.35 \mu\text{m}$  LDD NMOS uses error-function junctions ( $x_j = 0.170 \mu\text{m}$  metallurgical). Foundation checks (all at  $T = 300$  K): equilibrium Poisson (built-in  $+0.586/-0.44$  V, quadratic Newton); nonparabolic  $N_c = 2.857 \times 10^{19}$  (0.1% of the independent value; parabolic limit 0.09%

of closed form); bulk scattering engine detailed-balance  $2 \times 10^{-5}$ , zero-field  $T_e \rightarrow 309.7$  K, monotone heating. The DD solve at  $V_{gs} = V_{ds} = 3$  V gives the operating-point potential driving the SHE.

**Table 1:** Modelling taxonomy (Rev. 2 adds a *not-yet-converged* class).

| Element                                                 | Class                    | Basis / status                               |
|---------------------------------------------------------|--------------------------|----------------------------------------------|
| Method eqs., SHE reduction, moments                     | <b>paper</b>             | Liang <i>et al.</i>                          |
| Table-I params, spherical band                          | <b>paper</b>             | paper Sec. II                                |
| $L_{\text{eff}}, t_{ox}, x_j, N_D^\pm$                  | <b>paper</b>             | paper text                                   |
| 1-param Kane $\gamma(\varepsilon)$                      | <b>surrogate</b>         | paper: 2-band Fiegna/Brunetti                |
| LDD doping shape, mesh                                  | <b>reconstruction</b>    | ref. [7] unavailable                         |
| Channel $N_A, \Phi_{gate}, \tau_1, c_{op}, 1/\tau_{ii}$ | <b>surrogate</b>         | calibrated; disclosed                        |
| $G_{ii, \text{max}}$ magnitude                          | <b>not-yet-converged</b> | spatially unconverged; $5.24 \times 10^{26}$ |
| $T_e/G_{ii}$ peak offset                                | <b>not-yet-converged</b> | (coupled $60 \times 50$ ), still dropping    |
|                                                         |                          | unresolved at present mesh                   |

## Part II

# The 2-D SHE core, impact ionization, and self-consistency

## 4 Assembled operator, impact ionization, and moments

The solver discretizes the  $(\mathbf{r}, H)$  equation by box integration; contributions (spatial diffusion  $\times \Delta H$ , optical in/out  $\times A_{ij} \Delta H$ , acoustic SG flux  $\times A_{ij}$ ) share units and form a positive-diagonal system. Contacts inject a Maxwellian  $f_0 = n_{\text{contact}} e^{-\varepsilon/k_B T} / \int Z e^{-\varepsilon/k_B T} dH$ .

**Impact ionization in the electron BTE.** An event removes a primary electron at rate  $1/\tau_{ii}(\varepsilon)$  (Keldysh surrogate,  $\varepsilon_{th} = 1.1$  eV) and deposits two secondaries at  $\varepsilon' = (\varepsilon - \varepsilon_{th})/2$ , linearly distributed onto the neighbouring  $H$  nodes, with net electron gain +1 per event. This is *assembled into the electron collision operator* (primary depletion + secondary redistribution), so  $f_0$  is directly shaped by avalanche;  $G_{ii} = \int Z f_0 / \tau_{ii} d\varepsilon$  is a moment of that II-affected distribution. The absorbing  $\varepsilon_{max}$  cutoff acts on spatial, acoustic, and optical fluxes. Moments  $n, \langle \mathbf{v} \rangle, T_e, G_{ii}$ , and the terminal currents are reconstructed from the *same finite-volume face fluxes* as the operator.

## 5 Numerics: physics-split preconditioner

The large-grid system is solved by a multiplicative physics-split preconditioner. Because DOFs are numbered with  $H$  fastest, same-spatial-cell couplings (acoustic + optical + II redistribution + spatial diagonal) form contiguous local energy blocks — the **E block** — factored cheaply; the diagonal plus cross-cell couplings form the fixed- $H$  real-space **S block**. One application does  $E^{-1} \rightarrow S^{-1} \rightarrow E^{-1}$  with a fresh residual between stages; the solved matrix is never shifted (shifts touch only preconditioner factors). On the 328 544-DOF  $40 \times 34$  problem this solves to  $10^{-10}$  in  $\sim 15$  s (raw scaled residual  $1.22 \times 10^{-11}$ ), where the previous global ILU did not finish its setup in 120 s. The legacy path is retained as `method="legacy_ilu"` for regression.

## 6 Numerical audit of the one-way baseline

Table 2 is the extracted audit of the locked one-way reference (commit `f016e3c`); it is preserved as a regression benchmark.

**Table 2:** Clean  $40 \times 34$  numerical audit (impact ionization in the collision operator).

| Quantity                       | Value                                                    | Location / note             |
|--------------------------------|----------------------------------------------------------|-----------------------------|
| $n_{\max}$                     | $1.0516 \times 10^{20} \text{ cm}^{-3}$                  | source $n^+$ (0.082, 0.043) |
| $T_{e,\max}$                   | 3232.9 K                                                 | drain (0.595, 0.005)        |
| $G_{ii,\max}$                  | $1.3513 \times 10^{27} \text{ cm}^{-3} \text{ s}^{-1}$   | drain (0.595, 0.009)        |
| $ v _{\max}$                   | $9.732 \times 10^6 \text{ cm/s}$                         | drain surface (0.554, 0)    |
| $T_e - G_{ii}$ peak separation | 5 nm ( $\sim 1$ cell)                                    | co-located within mesh      |
| SHE currents $I_{src}/I_{drn}$ | $-1.7044/ + 1.7071 \times 10^{-4} \text{ A}/\mu\text{m}$ | mismatch 0.16%              |
| global continuity residual     | $-5.70 \times 10^{-6}$ (relative)                        | —                           |
| orphan states                  | 0                                                        | no artificial sinks         |
| $f_0$ pre-clip min             | $-3.3 \times 10^{-11}$                                   | 46/328544 tiny negatives    |

## 7 Grid refinement ( $40 \times 34$ vs $60 \times 50$ , one-way)

Refining the spatial mesh at fixed  $\Delta H = 12.5 \text{ meV}$ :

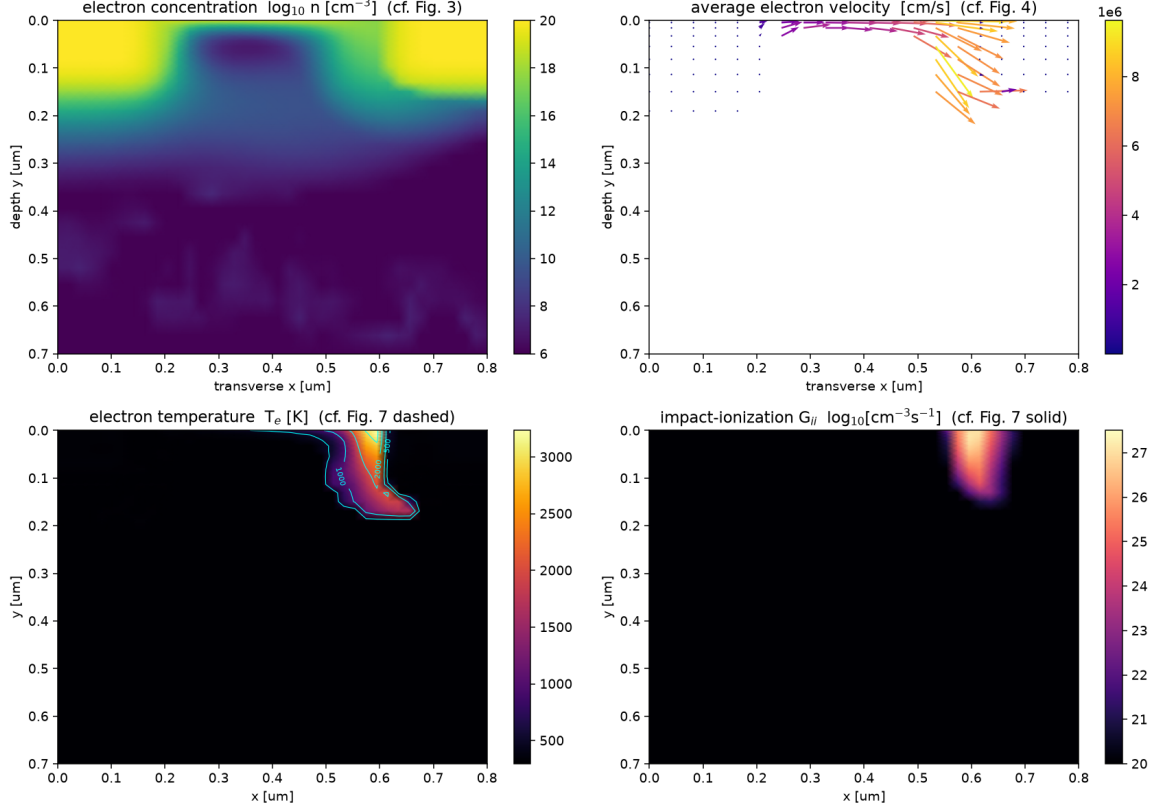
| Quantity                    | $40 \times 34$               | $60 \times 50$               | converged?              |
|-----------------------------|------------------------------|------------------------------|-------------------------|
| $n_{\max}$                  | $1.0516 \times 10^{20}$      | $1.0473 \times 10^{20}$      | yes ( $-0.4\%$ )        |
| $ v _{\max}$                | $9.73 \times 10^6$           | $10.0 \times 10^6$           | yes ( $+2.7\%$ )        |
| $T_{e,\max}$ [K]            | 3232.9                       | 3423.0                       | moderate ( $+5.9\%$ )   |
| $G_{ii,\max}$               | $1.3513 \times 10^{27}$      | $8.26 \times 10^{26}$        | <b>no</b> ( $-40\%$ )   |
| $T_e/G_{ii}$ separation     | 1 cell                       | 2 cells                      | grid-scale (unresolved) |
| current mismatch / residual | 0.16% / $5.7 \times 10^{-6}$ | 0.07% / $7.6 \times 10^{-6}$ | conserved               |

Transport ( $n$ ,  $|v|$ ,  $T_e$ ) is spatially robust;  $G_{ii}$  is *not* grid-converged (it depends on the high-energy tail); and the peak separation tracks the mesh, so it is unresolved, not a reproduced physical offset.

## 8 Self-consistent SHE $\leftrightarrow$ Poisson $\leftrightarrow$ hole loop

A damped outer iteration solves the SHE (with impact ionization) on the current potential, drives hole continuity with  $G_{ii}$  as pair generation (holding  $n_{\text{SHE}}$ ), updates Poisson from the SHE density, damps, and repeats;  $n$  is recomputed from the SHE after each potential update (not constitutively from DD). Over 12 iterations (Fig. 2) the maximum potential update falls monotonically  $17.7 \rightarrow 4.0 \text{ mV}$ , and the physical observables plateau:  $G_{ii,\max}$  is suppressed 37% ( $1.35 \rightarrow 0.85 \times 10^{27}$ ) and the drain current 29% ( $1.71 \rightarrow 1.21 \times 10^{-4} \text{ A}/\mu\text{m}$ ), while  $T_{e,\max}$  stays  $\sim 3240 \text{ K}$ . This is the physically expected negative feedback: the SHE electrons and the  $G_{ii}$ -generated holes reshape the potential, lowering the peak drain field and throttling avalanche. Final conservation is clean (relative number balance  $2.4 \times 10^{-5}$ , currents balance 0.14%).

**Honesty note.** The loop is *demonstrated and observably converging*, not converged to the prescribed tolerance: the plateaued observables are meaningful, but  $\max |\Delta\phi| \approx 4 \text{ mV}$  remains above the chosen 2 mV stopping criterion. Tighter convergence needs more iterations or reduced damping.



**Figure 1:** One-way SHE moments ( $40 \times 34$ ).  $n$  (Fig. 3), velocity field saturating at  $9.73 \times 10^6$  cm/s (Fig. 4; masked where  $n < 10^{13}$  cm $^{-3}$ ),  $T_e$  (Fig. 7 dashed, drain-localized),  $G_{ii}$  (Fig. 7 solid, drain-localized).

## 9 $G_{ii}$ convergence matrix and device recalibration

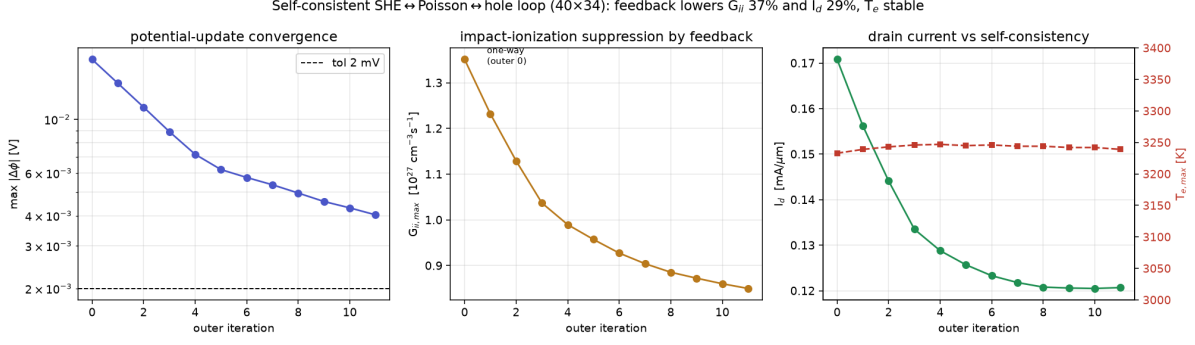
Combining one-way and coupled runs at both grids isolates how the impact-ionization magnitude responds to spatial refinement and to self-consistency (Table 3). The two effects are independent and *compound multiplicatively* ( $1.35 \times 10^{27} \times 0.61_{\text{grid}} \times 0.63_{\text{coupling}} = 5.2 \times 10^{26}$ ), and the coupled result is *still* spatially unconverged (coupled  $40 \times 34 \rightarrow 60 \times 50$ :  $-38\%$ , no plateau).

**Table 3:**  $G_{ii,\text{max}}$  [cm $^{-3}$ s $^{-1}$ ] over grid  $\times$  coupling.

|                | one-way               | self-consistent                         |
|----------------|-----------------------|-----------------------------------------|
| $40 \times 34$ | $1.35 \times 10^{27}$ | $8.50 \times 10^{26}$                   |
| $60 \times 50$ | $8.26 \times 10^{26}$ | <b><math>5.24 \times 10^{26}</math></b> |

An *energy* refinement to  $\Delta H = 6.25$  meV ( $E_{op} = 8$ ) moves  $G_{ii,\text{max}}$  only  $-2.4\%$  ( $T_e$   $-0.19\%$ ): the tail sensitivity is therefore **spatial** — resolution of the drain high-field/heating region — not energetic. No converged  $G_{ii}$  magnitude can be quoted; the robust results are transport and the ( $\sim 37\%$ , grid-independent) feedback fraction.

**Device recalibration.** A rigid gate offset  $\Phi_{gate} = -0.74$  V ( $dV_{th}/d\Phi_{gate} \approx +1$ ) recalibrates the extracted threshold to  $V_{th} = 0.614$  V (was 1.64 V); the  $V_g = 3$  V overdrive becomes 2.39 V. Re-running the coupled operating point on this device, then digitized Liang overlays, is the remaining quantitative work.



**Figure 2:** Self-consistent loop (40×34). Potential update converges  $17.7 \rightarrow 4.0$  mV; feedback suppresses  $G_{ii}$  37% and  $I_d$  29% (both plateauing) while  $T_e$  stays stable.

## Part III

# Assessment

## 10 Conclusions, classified

- **Core SHE transport:** validated and spatially robust for  $n$ ,  $|v|$ , and  $T_e$ .
- **Impact ionization:** now part of the assembled electron collision operator, with explicit primary depletion and secondary redistribution ( $G_{ii}$  is a moment of the II-affected  $f_0$ ).
- **Self-consistency:** SHE↔Poisson↔hole feedback *demonstrated and observably converging*; it materially suppresses avalanche generation and drain current while leaving  $T_e$  comparatively stable.
- **Conservation:** validated across one-way, refined, and coupled calculations (number balance  $10^{-5}$ – $10^{-7}$ ; terminal currents balance  $< 0.2\%$ ).
- **Numerics:** physics-split preconditioning removes the previous practical solver bottleneck.
- $G_{ii}$ : the earlier  $1.37 \times 10^{27}$  agreement with Liang was *fortuitous* (coarse, one-way). The coupled 60×50 run gives  $5.24 \times 10^{26} \text{ cm}^{-3} \text{ s}^{-1}$ , still  $-38\%$  below coupled 40×34 with no plateau, whereas energy refinement to 6.25 meV moves it only  $-2.4\%$ :  $G_{ii}$  is **spatially unconverged**, and *no converged magnitude can be quoted* (Table 3).
- $T_e/G_{ii}$  **offset:** unresolved at present mesh resolution; *no longer claimed reproduced*.
- **Device:** recalibrated to  $V_{th} = 0.614 \text{ V}$  ( $\Phi_{gate} = -0.74 \text{ V}$ ), replacing the earlier over-high 1.64 V.
- **Remaining quantitative gaps:** a spatially-converged  $G_{ii}$  (finer than 60×50), the coupled operating point re-run on the recalibrated device, and digitized Liang target-figure overlays.

## A Numerical and model specifications

$T = 300 \text{ K}$ , Boltzmann,  $m^* = 0.32m_0$ ,  $g_v = 6$ ,  $\alpha = 0.5 \text{ eV}^{-1}$ ,  $v_s = 9.04 \times 10^3 \text{ m/s}$ ; erfc doping per earlier revisions; SHE grid  $\Delta H = 12.5 \text{ meV}$  ( $E_{op}/\Delta H = 4$  integer),  $\varepsilon = 0$  reflecting /  $\varepsilon_{max}$  absorbing. Optical  $c_{op} = 4.69 \times 10^{-34} \text{ J m}^3/\text{s}$ ,  $\tau_1 = 10^{-14} \text{ s}/(\sqrt{\gamma}\gamma')$ ,  $\tau_{ac} = \tau_1$ ; impact ionization  $1/\tau_{ii} = P[(\varepsilon - \varepsilon_{th})/\varepsilon_{th}]^2 \Theta$ ,  $P = 2 \times 10^{13} \text{ 1/s}$ ,  $\varepsilon_{th} = 1.1 \text{ eV}$ , equal-split secondaries. Solve: physics-split E→S→E preconditioner, LGMRES tol  $10^{-10}$  (audit) /  $10^{-8}$  (coupled iterations). Coupled loop:  $\phi$ -damping 0.25,  $\leq 12$  outer iterations, 2 mV stop.

## B Provenance and audit history

Git commit 891e7c9; Python 3.13.14, NumPy 2.5.1, SciPy 1.18.0. Key commits: f016e3c (locked one-way audit), 7aad998 (60×50 refinement), 65461ad (physics-split solver), 047da93 (coupled loop). Four audit rounds fixed: over-claims, missing specs, junction geometry,  $N_c$  convergence, the reducible-collision zero-field- $T_e$  bug (acoustic operator), the global  $H$ -grid, the contact-Maxwellian normalization (report typo; code correct), boundary/velocity/current extraction, and finally delivered the real impact-ionization operator, the coupled driver, and the physics-split solver.

## References

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